Stat 517
Homework # 5

1. Let $X_1, X_2, \ldots, X_n$ be a random sample from a Gamma($\alpha = 3, \beta = \theta$) distribution, $0 < \theta < \infty$. Determine the MLE of θ.

Solution:

Step 1: Write down the likelihood. Since $X_i \sim$ Gamma($\alpha = 3$, $\beta = \theta$), the pdf of each observation is

$$f(x;\theta) = \frac{1}{\Gamma(3)\,\theta^3}\,x^2 e^{-x/\theta} = \frac{1}{2\theta^3}\,x^2 e^{-x/\theta}, \qquad x > 0,$$

using $\Gamma(3) = 2! = 2$. By independence, the likelihood function is

$$L(\theta) = \prod_{i=1}^n \frac{1}{2\theta^3}\,X_i^2 e^{-X_i/\theta} = \frac{1}{2^n\theta^{3n}}\left(\prod_{i=1}^n X_i^2\right)\exp\left(-\frac{1}{\theta}\sum_{i=1}^n X_i\right).$$

Step 2: Take the log.

$$\ell(\theta) = \log L(\theta) = -\,n\log 2 \;-\; 3n\log\theta \;+\; 2\sum_{i=1}^n \log X_i \;-\; \frac{1}{\theta}\sum_{i=1}^n X_i.$$

Note that only the second and fourth terms depend on θ.

Step 3: Differentiate and solve the score equation.

$$\ell'(\theta) = -\frac{3n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^n X_i.$$

Setting $\ell'(\theta) = 0$ and multiplying through by $\theta^2 > 0$,

$$-\,3n\theta + \sum_{i=1}^n X_i = 0 \quad\Longrightarrow\quad \theta = \frac{1}{3n}\sum_{i=1}^n X_i = \frac{\bar{X}}{3}.$$

Step 4: Verify that this is a maximum. The second derivative is

$$\ell''(\theta) = \frac{3n}{\theta^2} - \frac{2}{\theta^3}\sum_{i=1}^n X_i,$$

and evaluating at $\theta = \bar{X}/3$ (so that $\sum X_i = 3n\theta$),

$$\ell''\left(\frac{\bar{X}}{3}\right) = \frac{3n}{\theta^2} - \frac{2\,(3n\theta)}{\theta^3} = \frac{3n - 6n}{\theta^2} = -\,\frac{3n}{\theta^2} \;<\; 0.$$

Alternatively, $\ell'(\theta) > 0$ for $\theta < \bar{X}/3$ and $\ell'(\theta) < 0$ for $\theta > \bar{X}/3$, so the unique critical point is the global maximizer on $(0, \infty)$.

Therefore, the maximum likelihood estimator of θ is

$$\boxed{\hat{\theta} = \frac{\bar{X}}{3}}$$

2. Let $Y_1 < Y_2 < \cdots < Y_n$ be the order statistics of a random sample from a distribution with pdf $f(x;\theta)=1$, $\theta - \dfrac{1}{2} \le x \le \theta + \dfrac{1}{2}$, $-\infty < \theta < \infty$, zero elsewhere. This is a nonregular case. Show that every statistics $u(X_1, \ldots, X_n)$ such that

$$Y_n - \frac{1}{2} \le u(X_1, \ldots, X_n) \le Y_1 + \frac{1}{2}$$

is a mle of θ. In particular, $(4Y_1 + 2Y_n + 1)/6$, $(Y_1 + Y_n)/2$, and $(2Y_1 + 4Y_n - 1)/6$ are three such statistics. Thus, uniquess is not, in general, a property of mles.

Solution:

Step 1: Write the likelihood using indicator functions. Since $f(x;\theta) = 1$ on $\theta - \frac{1}{2} \le x \le \theta + \frac{1}{2}$ and zero elsewhere, the likelihood is

$$\begin{aligned} \mathcal{L}(\theta \mid x_1, \ldots, x_n) &= \prod_{i=1}^{n} I\big(\theta - \tfrac{1}{2} \le x_i \le \theta + \tfrac{1}{2}\big) \\ &= I\big(\theta - \tfrac{1}{2} \le y_1 \le \cdots \le y_n \le \theta + \tfrac{1}{2}\big), \end{aligned}$$

since all n constraints hold if and only if the smallest and largest observations satisfy them.

Step 2: Rewrite the constraint as an interval for θ. The event $\big\{\theta - \frac{1}{2} \le y_1 \text{ and } y_n \le \theta + \frac{1}{2}\big\}$ is equivalent to

$$y_n - \tfrac{1}{2} \;\le\; \theta \;\le\; y_1 + \tfrac{1}{2}.$$

Hence the likelihood, viewed as a function of θ, is the flat function

$$\mathcal{L}(\theta \mid x_1, \ldots, x_n) = \begin{cases} 1, & \theta \in \big[\, y_n - \frac{1}{2},\ y_1 + \frac{1}{2}\,\big], \\ 0, & \text{otherwise.} \end{cases}$$

This interval is nonempty with probability one: every observation lies in an interval of length 1, so $y_n - y_1 \le 1$, which gives $y_n - \frac{1}{2} \le y_1 + \frac{1}{2}$.

Step 3: Characterize all MLEs. The likelihood attains its maximum value 1 at every point of $\big[\, y_n - \frac{1}{2},\ y_1 + \frac{1}{2}\,\big]$ and is 0 outside. Therefore *every* statistic $u(X_1, \ldots, X_n)$ satisfying

$$Y_n - \tfrac{1}{2} \;\le\; u(X_1, \ldots, X_n) \;\le\; Y_1 + \tfrac{1}{2}$$

maximizes the likelihood and is thus an MLE of θ. In particular, the MLE is not unique.

Step 4: Verify the three particular statistics. Throughout, write $R = Y_n - Y_1$ and recall $0 \le R \le 1$.

(1) $u_1 = \dfrac{4Y_1 + 2Y_n + 1}{6}$. For the lower bound,

$$u_1 - \big(Y_n - \tfrac{1}{2}\big) = \frac{4Y_1 + 2Y_n + 1 - 6Y_n + 3}{6} = \frac{4(1 - R)}{6} \;\ge\; 0.$$

For the upper bound,

$$\left(Y_1 + \tfrac{1}{2}\right) - u_1 = \frac{6Y_1 + 3 - 4Y_1 - 2Y_n - 1}{6} = \frac{2\,(1-R)}{6} \;\geq\; 0.$$

(2) $u_2 = \dfrac{Y_1 + Y_n}{2}$ (the midrange).

$$u_2 - \left(Y_n - \tfrac{1}{2}\right) = \frac{Y_1 - Y_n + 1}{2} = \frac{1-R}{2} \;\geq\; 0, \qquad \left(Y_1 + \tfrac{1}{2}\right) - u_2 = \frac{1-R}{2} \;\geq\; 0.$$

(3) $u_3 = \dfrac{2Y_1 + 4Y_n - 1}{6}$.

$$u_3 - \left(Y_n - \tfrac{1}{2}\right) = \frac{2Y_1 + 4Y_n - 1 - 6Y_n + 3}{6} = \frac{2\,(1-R)}{6} \;\geq\; 0,$$

$$\left(Y_1 + \tfrac{1}{2}\right) - u_3 = \frac{6Y_1 + 3 - 2Y_1 - 4Y_n + 1}{6} = \frac{4\,(1-R)}{6} \;\geq\; 0.$$

In each case both differences are nonnegative because $R \leq 1$, so all three statistics lie in $\left[\,Y_n - \tfrac{1}{2},\, Y_1 + \tfrac{1}{2}\,\right]$ and are MLEs of θ. Note that u_1, u_2, u_3 are genuinely different statistics (they coincide only when $R = 1$), so uniqueness is not, in general, a property of maximum likelihood estimators.

3. Suppose $X_1, \ldots, X_n$ are iid with pdf $f(x;\theta) = 2x/\theta^2$, $0 < x \le \theta$, zero elsewhere. Note this is a nonregular case. Find:

   (a) The mle of $\hat{\theta}$ for θ.

   (b) The constant c so that $E\left(c\hat{\theta}\right) = \theta$.

   (c) The mle for the median of the distribution. Show that it is a consistent estimator.

Solution:

(a) The MLE of θ. The likelihood function is

$$\begin{aligned}\mathcal{L}(\theta \mid x_1, \ldots, x_n) &= \prod_{i=1}^{n} \frac{2x_i}{\theta^2} I(0 < x_i \le \theta) \\ &= \frac{2^n \prod_{i=1}^{n} x_i}{\theta^{2n}} I(0 < y_1 \le \cdots \le y_n \le \theta) = \frac{2^n \prod_{i=1}^{n} x_i}{\theta^{2n}} I(y_n \le \theta),\end{aligned}$$

where $y_1 \le \cdots \le y_n$ are the ordered observations: all n constraints $x_i \le \theta$ hold if and only if the largest one does.

As a function of θ, the likelihood is zero for $\theta < y_n$ and proportional to θ^{-2n} for $\theta \ge y_n$, which is strictly decreasing. Hence the likelihood is maximized at the smallest admissible value of θ, namely

$$\boxed{\hat{\theta} = Y_n = \max\{X_1, \ldots, X_n\}.}$$

Note this is a nonregular problem (the support depends on θ), so the MLE is found by inspection of the likelihood rather than by solving $\ell'(\theta) = 0$; here the MLE is unique.

(b) The unbiasing constant c.

Step 1: Distribution of Y_n. For $0 < x \le \theta$, the cdf of a single observation is

$$G(x) = \int_0^x \frac{2t}{\theta^2}\, dt = \left(\frac{x}{\theta}\right)^2.$$

By independence, for $0 < y \le \theta$,

$$F_{Y_n}(y) = P(Y_n \le y) = \prod_{i=1}^{n} P(X_i \le y) = \left(\frac{y}{\theta}\right)^{2n}, \qquad f_{Y_n}(y) = \frac{2n\, y^{2n-1}}{\theta^{2n}}.$$

Step 2: Compute $E(Y_n)$.

$$E(Y_n) = \int_0^\theta y\, \frac{2n\, y^{2n-1}}{\theta^{2n}}\, dy = \frac{2n}{\theta^{2n}} \int_0^\theta y^{2n}\, dy = \frac{2n}{\theta^{2n}} \cdot \frac{\theta^{2n+1}}{2n+1} = \frac{2n}{2n+1}\theta.$$

Thus $E(c\hat{\theta}) = c\,\frac{2n}{2n+1}\,\theta = \theta$ requires

$$\boxed{c = \frac{2n+1}{2n}.}$$

(As expected, $\hat{\theta} = Y_n$ underestimates θ since $Y_n \le \theta$ always, and $c > 1$ corrects this bias.)

(c) The MLE of the median and its consistency.

Step 1: The median as a function of θ. Setting $G(m) = 1/2$,

$$\left(\frac{m}{\theta}\right)^2 = \frac{1}{2} \quad \Longrightarrow \quad m = m(\theta) = \frac{\theta}{\sqrt{2}}.$$

Step 2: Apply the invariance property of MLEs. Since $m(\theta) = \theta/\sqrt{2}$ is a (one-to-one) function of θ, the invariance property of maximum likelihood estimation gives

$$\boxed{\hat{m} = m(\hat{\theta}) = \frac{Y_n}{\sqrt{2}}.}$$

Step 3: Consistency of Y_n. We show that

$$Y_n \xrightarrow{p} \theta.$$

Since every observation satisfies $0 < X_i \le \theta$, we have

$$Y_n \le \theta.$$

Thus, for any $0 < \epsilon < \theta$,

$$\begin{aligned} P\left(|Y_n - \theta| > \epsilon\right) &= P\left(\theta - Y_n > \epsilon\right) \\ &= P\left(Y_n < \theta - \epsilon\right). \end{aligned}$$

Using the cdf of Y_n,

$$P(Y_n \le y) = \left(\frac{y}{\theta}\right)^{2n}, \qquad 0 < y \le \theta,$$

we obtain

$$P\left(|Y_n - \theta| > \epsilon\right) = \left(\frac{\theta - \epsilon}{\theta}\right)^{2n}.$$

Because

$$0\frac{\theta - \epsilon}{\theta} < 1,$$

it follows that

$$\left(\frac{\theta - \epsilon}{\theta}\right)^{2n} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

For $\epsilon \ge \theta$, the probability is zero because $0 < Y_n \le \theta$. Therefore,

$$\boxed{Y_n \xrightarrow{p} \theta.}$$

Step 4: Consistency of the median estimator. The median of the distribution is

$$m(\theta) = \frac{\theta}{\sqrt{2}},$$

and its MLE is

$$\hat{m} = \frac{Y_n}{\sqrt{2}}.$$

Since

$$Y_n \xrightarrow{p} \theta$$

and the function

$$g(t) = \frac{t}{\sqrt{2}}$$

is continuous, the continuous mapping theorem gives

$$g(Y_n) \xrightarrow{p} g(\theta).$$

Therefore,

$$\frac{Y_n}{\sqrt{2}} \xrightarrow{p} \frac{\theta}{\sqrt{2}},$$

or equivalently,

$$\boxed{\hat{m} \xrightarrow{p} m(\theta).}$$

Hence $\hat{m} = Y_n/\sqrt{2}$ is a consistent estimator of the median.

4. Suppose $X_1, X_2, \ldots, X_n$ are iid with pdf $f(x;\theta) = \theta^{-1}e^{-x/\theta}$, $0 < x < \infty$, zero elsewhere. Find the mle of $P(X \le 2)$ and show that it is consistent.

Solution:

Step 1: Find the MLE of θ. The likelihood function is

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} \theta^{-1}e^{-x_i/\theta} = \theta^{-n}\exp\left(-\frac{1}{\theta}\sum_{i=1}^{n} x_i\right),$$

so the loglikelihood is

$$\ell(\theta) = -n\log\theta - \frac{1}{\theta}\sum_{i=1}^{n} x_i.$$

Differentiating with respect to θ,

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} x_i.$$

Setting $\ell'(\theta) = 0$ and multiplying through by $\theta^2 > 0$,

$$-n\theta + \sum_{i=1}^{n} x_i = 0 \quad \Longrightarrow \quad \theta = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}.$$

Step 2: Verify that this is a maximum. The second derivative is

$$\ell''(\theta) = \frac{n}{\theta^2} - \frac{2}{\theta^3}\sum_{i=1}^{n} x_i,$$

and evaluating at $\theta = \bar{x}$ (so that $\sum x_i = n\bar{x} = n\theta$),

$$\ell''(\bar{x}) = \frac{n}{\bar{x}^2} - \frac{2n\bar{x}}{\bar{x}^3} = -\frac{n}{\bar{x}^2} < 0.$$

Hence the unique critical point is the global maximizer, and

$$\boxed{\hat{\theta} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.}$$

Step 3: Express $P(X \le 2)$ as a function of θ.

$$p(\theta) := P(X \le 2) = \int_0^2 \theta^{-1}e^{-x/\theta}\,dx = \left[-e^{-x/\theta}\right]_0^2 = 1 - e^{-2/\theta}.$$

Step 4: Apply the invariance property of MLEs. Since $p(\theta) = 1 - e^{-2/\theta}$ is a function of θ, the invariance property of maximum likelihood estimation gives

$$\boxed{\hat{p} = p(\hat{\theta}) = 1 - e^{-2/\bar{X}}.}$$

Step 5: Consistency of $\hat{\theta}$. We show that

$$\hat{\theta} \xrightarrow{p} \theta.$$

Since $X_1, \ldots, X_n$ are iid with

$$E\left(X_i\right) = \theta < \infty,$$

the Weak Law of Large Numbers gives

$$\hat{\theta} = \bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i \xrightarrow{p} E\left(X_1\right) = \theta.$$

Therefore,

$$\boxed{\hat{\theta} \xrightarrow{p} \theta.}$$

Step 6: Consistency of $\hat{p}$. The function

$$g(s) = 1 - e^{-2/s}$$

is continuous on $s > 0$. Since

$$\hat{\theta} \xrightarrow{p} \theta$$

and g is continuous at $\theta > 0$, the continuous mapping theorem gives

$$g\left(\hat{\theta}\right) \xrightarrow{p} g\left(\theta\right).$$

Therefore,

$$1 - e^{-2/\bar{X}} \xrightarrow{p} 1 - e^{-2/\theta},$$

or equivalently,

$$\boxed{\hat{p} \xrightarrow{p} P\left(X \leq 2\right).}$$

Hence $\hat{p} = 1 - e^{-2/\bar{X}}$ is a consistent estimator of $P(X \leq 2)$.